

Taijitu Induction Generator (TIG): A Contactless, Vacuum-Compatible, High-Power Induction Generator with Opposing Vortex Fields for Space Propulsion and MULAN Integration

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Abstract

We present a novel contactless induction generator based on a rapidly rotating Taijitu (Yin-Yang) permanent magnet rotor, suspended via magnetic levitation in a vacuum environment. The geometry consists of two oppositely polarized half-disks that produce counter-rotating magnetic vortices. This configuration enables two independent induction coils to capture voltages with opposite phases, which when combined in series deliver up to four times the power of a conventional single-dipole generator of equal size. The rotor is encapsulated in a water-ice shell that provides thermal stabilization, structural integrity, and radiation shielding. The absence of mechanical bearings, brushes, and atmospheric drag allows sustained operation at angular velocities exceeding 10 000 rad/s, yielding specific power densities on the order of megawatts per kilogram. This generator is designed to power the MULAN propellantless thruster, solving its fundamental current limitation. We discuss the physical principles, materials selection, thermal management, and expected performance for space missions lasting centuries.

1 Introduction

Conventional electric generators rely on mechanical bearings, brushes, and atmospheric cooling, all of which introduce friction, wear, and limited operational lifetimes. In space, the absence of gravity and atmosphere offers unique opportunities for contactless, friction-free rotating machinery. This work proposes a generator that exploits these conditions by combining three key innovations:

1. **Taijitu (Yin-Yang) rotor geometry** – two oppositely polarized magnetic half-disks that generate opposing magnetic vortices.
2. **Magnetic levitation** – the rotor is suspended without physical contact, eliminating bearing wear.

3. **Water-ice encapsulation** – the rotor is embedded in a frozen water shell that provides thermal mass, structural rigidity, and radiation shielding.

The resulting device, termed the **Taijitu Induction Generator (TIG)**, delivers unprecedented specific power (MW/kg) and operational lifetime (centuries) in deep-space conditions.

2 Physical Principle

2.1 Magnetic Field Configuration

The rotor consists of two semicircular permanent magnets with magnetizations:

$$\mathbf{M}_1 = +M_0\hat{z}, \quad \mathbf{M}_2 = -M_0\hat{z},$$

arranged in a Yin-Yang pattern. The magnetic dipole moments are:

$$\mathbf{m}_1(t) = m_0(\cos\omega t\hat{x} + \sin\omega t\hat{y}), \quad \mathbf{m}_2(t) = -m_0(\cos\omega t\hat{x} + \sin\omega t\hat{y}).$$

2.2 Opposing Vortex Fields

When the rotor spins with angular velocity ω , each half-disk generates a time-varying magnetic field. Due to the opposite polarizations, the induced electric fields in the surrounding space have opposite phase:

$$\mathbf{E}_1(t) = \mathbf{E}_0 \cos(\omega t + \phi), \quad \mathbf{E}_2(t) = -\mathbf{E}_0 \cos(\omega t + \phi).$$

2.3 Dual-Coil Induction

Two stationary pickup coils (Coil A around the first half-disk, Coil B around the second) experience voltages:

$$V_A(t) = V_0 \sin(\omega t + \phi), \quad V_B(t) = -V_0 \sin(\omega t + \phi).$$

Connecting the coils in series yields:

$$V_{\text{total}}(t) = V_A(t) - V_B(t) = 2V_0 \sin(\omega t + \phi).$$

2.4 Power Enhancement

The peak power delivered to a load resistance R is:

$$P_{\text{single}} = \frac{V_0^2}{R}, \quad P_{\text{TIG}} = \frac{(2V_0)^2}{R} = 4P_{\text{single}}.$$

Thus, the Taijitu geometry quadruples the power output compared to a conventional single-dipole generator of identical size and rotation speed.

3 Engineering Design

3.1 Rotor Construction

- **Magnetic material:** Samarium-Cobalt (SmCo) or Neodymium-Iron-Boron (Nd-FeB) with protective coating. SmCo is preferred for century-scale stability (<1% demagnetization per century).
- **Structural core:** Carbon-fiber composite (density 1.6 g/cm³, tensile strength >3 GPa) to withstand centrifugal forces.
- **Encapsulation:** The rotor is encased in a 2–5 mm layer of pure water, which freezes in vacuum. The ice shell provides:
 - Thermal inertia (smooths temperature fluctuations)
 - Structural integrity (holds the magnet in place)
 - Radiation shielding (stops energetic particles)
 - Self-healing (sublimated ice can be replenished from a small water reservoir)

3.2 Magnetic Levitation (Bearingless Suspension)

The rotor is suspended by an array of stationary electromagnetic coils that generate a restoring force:

$$F_{\text{lev}} = \frac{\mu_0 m_1 m_2}{2\pi d^4},$$

where d is the gap between rotor and stator. In microgravity, only minimal force is required to center the rotor. Active feedback control maintains stable levitation.

3.3 Stator and Pickup Coils

- Two independent copper or high-temperature superconducting (HTS) coils surround the rotor.
- Coils are cooled by the ambient deep-space temperature (≈ 20 K) when radiatively shielded from the Sun.
- HTS coils (e.g., YBCO or BSCCO) operate with near-zero resistance, eliminating ohmic losses.

3.4 Rotation Drive (Torque without Contact)

The rotor is accelerated by a rotating magnetic field generated by a separate set of stator windings (similar to an induction motor). Once at operating speed, minimal torque is needed to overcome residual eddy-current drag.

Table 1: Design parameters for a 100 MW TIG unit.

Parameter	Value
Rotor radius	0.5 m
Magnetic moment (m_0)	100 A·m ²
Angular velocity (ω)	10 000 rad/s ($\approx$ 95 000 RPM)
Coil radius	0.6 m
Number of turns per coil	5000
Load resistance	0.01 Ω (HTS)

4 Expected Performance

4.1 Baseline Parameters (Space-Optimized)

4.2 Output Power

Using the dual-coil series configuration:

$$V_{\text{peak}} = \omega \cdot B_0 \cdot A_{\text{coil}} \cdot N_{\text{coils}} \cdot 2,$$

$$P_{\text{rms}} = \frac{V_{\text{peak}}^2}{2R_{\text{load}}}.$$

For the parameters above:

$$B_0 \approx 2 \times 10^{-4} \text{ T}, \quad \Phi_{\text{peak}} \approx 0.4 \text{ Wb}, \quad V_{\text{peak}} \approx 8000 \text{ V}, \quad P_{\text{rms}} \approx 128 \text{ MW}.$$

4.3 Specific Power

Rotor mass: ≈ 10 kg (composite + magnet) + ice shell (≈ 50 kg). Stator mass: ≈ 100 kg (HTS coils, frame, shielding). Total mass: ≈ 160 kg. Specific power: ≈ 0.8 MW/kg – **four orders of magnitude higher than solar panels or nuclear reactors**.

4.4 Operational Lifetime

- No mechanical contact $\rightarrow$ zero wear.
- SmCo magnets lose $<1\%$ magnetization per century.
- HTS coils do not degrade at 20 K.
- Ice layer sublimates at $\approx 10^{-6}$ kg/m²/s in vacuum; a 1-ton water reservoir replenishes the shell for >1000 years.
- Estimated continuous operation: **300–1000 years** without major maintenance.

5 Integration with MULAN Thruster

The MULAN (Momentum Uplifting via Light Amplification in Nanoconfinement) thruster requires high electrical power to drive a laser, which then produces thrust via $F = P_{\text{opt}}K/c$.

5.1 Power-to-Thrust Conversion

Given $P_{\text{el}} = 128$ MW, laser efficiency $\eta_{\text{laser}} = 0.5$, $K = 15$, $\eta_{\text{cascade}} = 0.8$:

$$P_{\text{opt}} = 64 \text{ MW}, \quad F = \frac{64 \times 10^6 \cdot 15 \cdot 0.8}{3 \times 10^8} \approx 2.56 \text{ N}.$$

A single TIG unit thus delivers 2.56 N of propellantless thrust – sufficient for attitude control, orbit maintenance, and slow interplanetary transfers for spacecraft up to ~ 100 tons.

5.2 Modular Scaling

Multiple TIG units can be combined:

Table 2: Modular scaling for different spacecraft masses.

# Units	Total power	Thrust	Applicable spacecraft mass
1	128 MW	2.56 N	10–100 t
4	512 MW	10.2 N	100–500 t
10	1.28 GW	25.6 N	500–2000 t
100	12.8 GW	256 N	2000–10 000 t

6 Manufacturing and Testing Roadmap

6.1 Earth Prototype (1 kW)

1. Fabricate a 5 cm diameter SmCo rotor with water-ice shell.
2. Suspend using active magnetic bearings in a vacuum chamber.
3. Drive rotation via external stator (brushless DC motor principle).
4. Measure induced voltage in dual coils, verify $4\times$ power enhancement.

6.2 Space Prototype (100 kW)

1. Scale to 20 cm diameter rotor.
2. Replace copper coils with HTS (YBCO) cooled by passive radiator to 20 K.
3. Launch to LEO, demonstrate operation in microgravity.
4. Connect to a small MULAN chip ($P_{\text{el}} = 100$ kW), measure thrust.

6.3 Full-Scale Space Unit (100 MW)

1. 1 m diameter rotor, SmCo magnets, carbon-fiber reinforcement.
2. HTS stator with active magnetic levitation.
3. Integral water reservoir (1 ton) for ice replenishment.
4. Designed for 300+ year continuous operation.

7 Conclusion

The Taijitu Induction Generator (TIG) represents a paradigm shift in space power generation. By exploiting the unique geometry of opposing magnetic half-disks, dual-coil phase opposition, vacuum operation, and magnetic levitation, it achieves specific power densities exceeding 0.8 MW/kg and operational lifetimes measured in centuries. When coupled with the MULAN propellantless thruster, it enables sustained, fuel-free propulsion for deep-space missions, generation ships, and long-term orbital infrastructure. The technology is within reach of current materials science and can be validated with a modest Earth-based prototype.

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